

Patrick ASIEDU BSc MSc CEng MICE FCIHT Eurlng.

Technical Director

PROFILE

Patrick is a Professional Chartered Engineer (CEng) with over 29 years wide range of experience in the following competences:

- Providing Technical advice / Managing Engineering assurance.
- checking & reviewing detailed design. Managing budget and invoicing
- Ensuring compliance with standards, legislations, contract requirements and specifications and carrying out audit
- Design Management, Coordinating design, and technical inputs at interdisciplinary design meeting & Client Presentation
- Liaising and proactive engagement with clients, contractors & project team including issuing Early Warning Notice
- Preparing Tenders Summary. Delivering multidisciplinary projects on time and to budget
- Planning, undertaking and managing inspection for assessment of railway infrastructures including bridges.
- Supervising heavy Civil Engineering works and checking compliance on site
- Survey Management, carrying out Monitoring, Setting out and Topographic survey with total station (EDM)
- Preparing Environmental Impact assessment, Designers / Construction Risk Assessment & Method Statements, Technical Workscope and Projects reporting
- Engineering Design using Staad.Pro, Mathcad, CS TED, Ring 3.0 H, Strand 7, CAD applications.
Excellent proficiency in Microsoft office tools

CORE SKILLS

- Civil & Structural Engineering Design Management
- Temporary Works Design, Coordination & Management
- CRE (Civils / Structures Design / Construction)
- Project Engineering
- Commercial Management
- Bid submission & Tendering
- Project Finance & Control
- Contract Management
- Environmental Management
- Works Packages Management
- Health & Safety Management

PROFESSIONAL EXPERIENCE

❖ TIFAND LIMITED — *(Mai 2024 – Present)*

Technical Director

- Leading temporary works design and checking, integrated with permanent works design as applicable
- Providing tender and bid winning support including outline design schemes, feasibility studies and value engineering solutions
- Managing technical aspects of projects, ensuring compliance with CDM Regulations and TIFAND's Design & Quality Management System
- Developing positive relationships with clients and design teams, representing TIFAND at project meetings
- Mentoring and developing staff within TIFAND Design teams, promoting the engineering profession and maintaining knowledge transfer

❖ BAM Nuttall – TransPennine Route Upgrade (TRU) — *(April 2020 – March 2024)*

Engineering Manager CRE (Civils / Structures / Construction)

Reviewed I reviewed design and managed Engineering assurance process for the enabling and Installation works. Issue Review and workflow TQ and keep up to date records. Attended IDC/IDR meetings. Reviewed and signed WPPs, ITPs, etc. Ensured constructions work complied with design. Issued Form E. Mentored CRE delegates. Reviewed Form E issued by Delegated CREs.

- Ensured all civil engineering work complies with Network Rail standards, Eurocodes, and relevant legislation (such as the CDM Regulations 2015).
- Reviewed the design for buildability and ensuring the site team understands the design intent.
- Managed and resolved technical queries between the site team and the designer when site conditions differ from the plan.
- Reviewed and signed off on WPPs and Task Briefing Sheets (TBS) to ensure the engineering methodology is safe and effective.
- Reviewed and approved ITPs, ensuring that all necessary checks and "Hold Points" are identified and signed off during construction.
- Coordinated with the Temporary Works Coordinator (TWC) to ensure that temporary structures (like scaffolding or cofferdams) do not compromise the permanent works or rail safety.
- Carried out regular site visits to verify that work is being delivered according to the approved drawings and specifications.
- Identified, recorded, and resolved any non-conformances in the civils works.
- Participated in Inter-Disciplinary Checks (IDC) meetings to ensure civils work doesn't clash with other disciplines like Signaling, OLE (Overhead Line Equipment), or Track
- Acted as a technical mentor for site engineers and section engineers.

❖ AMEY INFRASTRUCTURE WALES / AMEY RAILS — *(May 2018 – March 2020)*

Temporary Works Design Manager & Principal Project Engineer

- Reviewed and accepted designs on behalf of Transport for Wales
- Ensured Civil designs are consistent with scope of Project and are buildable
- Liaised with design house to ensure their requirement are effectively communicated
- Ensured all temporary works onsite are managed in accordance with temporary works procedure, standards, and legislations
- Checked Civil & Structural Engineering designs and calculations
- Review design and issue DRN
- Coordinated engineering review team
- Reviewed DRN response and liaise with DPE for acceptance
- Ensure design complies with CR-T, relevant HSE legislation including CDM
- Responded to queries from design house and contractors
- Ensured temporary works procedures are followed. established, maintained, and updated
- Established and maintain Temporary Works Register
- Prepared design briefs and ensured appropriate temporary works designs are in place and complied with relevant design briefs
- Ensure Temporary Works Supervisors (TWSs) are onboarded on the project and briefed on Temporary Works Procedures and Permit Issue system.

❖ ALSTOM TRANSPORT — *(June 2015 – July 2018)*

Contractor Responsible Engineer (CRE Design & Construction)

- Managed Civil Engineering Design & Assurance and Providing Technical advice
- Liaised and provided required information to client, subcontractors and stakeholders
- Raised and resolved technical queries and early warning notice
- Reviewed designs and associated documentation, WPPs, TBSs & ITPs, and challenged standards
- Carried out quality assurance checks to ascertain designs to be submitted are efficient/economic and safe design to build
- Ensured construction is carried out in accordance with designs
- Work package management, Verifying contractors claims and provided information to Quantity Surveyors (Qs) to approve and challenge claims.
- Reviewed designs to ensure compliance and constructability
- Raised and reviewed TQ's, Field Change Notice/Request (FCN/ FCR) and Non-Conformity Reports (NCR)
- Issued Form 5 for cable bridges, portal gantries and location cases
- Reported construction progress / issued documentation to the CEM , Commercial Manager and Project Director
- Identified Engineering Change and compiled information on changes of scope/preferential design
- Managed closing out of civil/ structural aspect of project
- Liaised with Network Rail DPE's and project team
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Civil Engineering Manager (Civil & Structural Design)

- Provided civil and structural engineering advice and design support
- Reviewed and managed designs from Network Rail GRIP 1 – 5
- Produced detailed designs including signalling foundations, retention designs and piles for overhead signals equipment, etc.

❖ BALFOUR BEATTY RAIL — *(February 2013 – May 2015)*

Principal Engineer & Contractor Responsible Engineer (**CRE Civils**)

- Lead and managed design delivery and assurance
- Reviewed designs and provided technical support on design of Bridges and refurbishment/renewal schemes, temporary works for crane working platforms
- Carried out estimating and tendering for major enabling works
- Liaised with Client and NetworkRail DPEs and representing design team at Interdisciplinary Design Coordination and Review meetings

❖ SITA SERC — *(December 2010 – January 2013)*

Design Manager

Manage and delivered multidiscipline design comprising drainage, bridge Assessment and strengthening, structural and formation designs, CAD operation and surveys for the construction of Severnside Energy Recovery Centre at Avonmouth. Responded to queries, carried out peer review and ensured design Quality Assurance. Liaise with Network Rail and gave early warning notice to ensure efficient delivery. Recorded changes requested for Quantity surveyors to pursue claim.

❖ SUPPLEMENTARY KEY EXPERIENCE

Project Engineer / Lead Design Engineer - Thameslink Enabling Work Project (£1.9M)

I compiled documentation of team experience of previous structural assessment, letters of commendation, and business capabilities to support our tender. I successfully managed a multidisciplinary team to inspect / survey, supervised site investigation work, carried out gantry assessment, designed cable route and structural modification schemes and delivered the project in accordance with programme / budget. I liaised with client, answered queries and attended design coordination/review meetings.

Projects include – Assessment/ modification to Cable bridges /Gantries, reinforced concrete slab for Landsman Way RRAP, London Bridge equipment room opening, cable support bridge and cable routes design

Project Engineer - Reconstruction of Bridges 81A and 81B, for London Underground (£1.4M)

I delivered design, coordinated senior engineering team and liaised with client. Led a team of 8 personnel to carry out category III design check of cable diversion Bridge, undertook structural condition (dilapidation) survey and produced report. carried out and supervised survey, setting out and monitoring work. I outsourced and managed soil investigation works. I designed temporary trestles and foundation to support bridge superstructure.

Survey Manager - Topographic survey (£216k)

I have successfully led/managed a team of 5 Engineers and chainmen to carry out topographic surveys and processed data / drawings to facilitate strengthening and design schemes. As a multi-skilled Engineer, I also carry out reconnaissance survey and priced topographic surveys whenever these services are required during lean periods. *Projects delivered include Parsons Brinkerhoff Kent Embankment Topographic survey, London Victoria station (Eccleston, Elizabeth and Ebury Bridges), Birmingham Moor Street and Snow Hill tunnel and Bexhill as-built survey. Long John Hill Bridge Survey for Atkins Rail on the Angelia Route Collaboration survey.*

Project Engineer - Whitechapel Station Monitoring Works (£175k)

I led a team of 3 engineers/ surveyors to carry out a monitoring regime using state of the art Amberg GRP 3000 technology with total station to record detrimental effects from tunnelling and deep excavation.

Senior Engineer – Closed Branch Line (CBL) Project for British Rail Board (BRB) Residuary Ltd (£1.24M)

The £250K a year project comprises structural assessment and examination, inspection and investigation work. I led a team of 3 Engineers to carry out inspection/assessment of structures and submitted report for Lune valley Viaduct, Lawrence hill Viaduct -Bristol, Dalston Lane Tunnel – London.

Design Manager - Hackney Station Booking Hall (£19k)

Managed / led a team of 3 Engineers to design temporary work live load trestles to support Kirow 810 and 250 cranes.

Senior Engineer - Ore Valley Wall Design & Selhurst Traincare Depot Project (£34k)

Undertaking category III check on Ore valley wall design and checking rail crossing design for Selhurst depot.

Senior Engineer - Northfleet Reconnection Project (115k)

Designed, checked and supervised multi-discipline design scheme to install new railway sidings. Checked structural assessments of existing structures, designed canopy and access stair for control room, new signal posts, duct routes.

King Norton Footbridge Design

Used STAAD.pro design software to carry out structural analysis of Kings Norton Footbridge. Designed bolt connections to facilitate transportation of beams. Checked designs and reviewed submissions.

Duct Road Bridge and River Lockstock Bridge Designs

Modelling of superstructure and Structural analysis for Category III Design Checking

Design Manager – Design & Strengthen Project (£700k)

I successfully managed a team of 8 Engineers and a CAD technician to deliver bridge strengthening, structural designs, drainage and earth retention scheme designs for B&M Mchugh. I liaised with client B&M, Network Rail and responded to technical queries. Projects delivered include:

Mechanical Viaduct - Deptford

Designed steel grillage encased in concrete to renew padstone supporting platform extension and prepared technical workscope. The innovative design reduced heavy craning and working at height.

High Street Footbridge - Gray's station

Undertaking a category III check – Assessing and checking designed temporary support system to existing Grays Station footbridge trestles and flights. Designed new Staircases using GRP to eliminate corrosion.

Homerton Station Toe retention scheme

Designed embankment and toe retaining wall incorporating “groundomat” pile innovation with miniature pile rig. Designed drainage work and new boundary fencing involving the installation of general purpose palisade fence. I was the Site Engineer for the emergency construction work.

Whittlesea Dyke Bridge

Designed grid using parallel flange channels (PFC) and Pattress plates (transverse and longitudinal) to tie the separated and flexing quoin to the remaining (greater mass) L/E abutment. My temporary solution strengthened the flexing Whittlesea Dyke Underbridge abutment to enable the lifting of an emergency speed restriction (ESR) of 20mph imposed on the Down line (causing delays) whilst I developed a more permanent solution comprising of underpinning and stitching of the fractures abutment. I incorporated the tie bar system of the temporary solution into the permanent solution. I was the resident engineer for the installation works

Other projects include: - Mutton Hall Underbridge ballast retention schemes and Tonford -A bridge Grillage design.

Falmer Station Footbridge

I planned and supervised the site survey and carried out the load analysis of the structure for the lifting operation. I also checked the final design calculations.

Design Manager - Design & Strengthen Project (£175K) for Dyer and Butler

I tendered and worn design works. Managed and delivered strengthening work design for 17 structures to budget and programme. I was the single point of contact with the clients- Dyer and Butler / Network Rail. Projects include:

Fairfield Street

Designed a durable steel grillage to restrain wheel timber, resist overturning forces due to nosing and transfer load from wheeltimber to the abutment. The grillage also reduced heavy tamping during the application of live loads from trains to acceptable levels.

Woodlands Way

Proposed design concept, supervised and checked cost effective innovative underpinning scheme comprising micro piles (offset from wingwalls) and steel cantilever plates to support wingwall instead of demolishing and rebuilding. Design minimised demolition, installation and cable strike risk and reduced disruption to members of public.

Design Engineer 2005-07/ Senior Engineer 2008-10 – Structural Assessment Contract-2005-2010 (£40M)

I was the business development manager for Survey and structural Assessment. I won work, led six Assessment Engineers in undertaking and delivering assessment of bridges on time and to budget and ensuring compliance of relevant codes. I inspected and determined the structural capacity of bridges and recommended strengthening schemes. I checked assessment calculations, reports and attended validation and verification meeting with client.

Structures assessed include London Bridge Station Footbridge, East Street Underbridge and Earlswood Station Subway.

Project Engineer - Tunnel Investigation Works

I managed 4 contractors undertaking site investigations including Ground Penetration Radar, coring, surveying and scanning and condition surveys at St John Wood & Yelverton Tunnels and submitted investigation report to clients.

Design Engineer– Structural Examination

I examined station structures including footbridges and produced defect reports with recommendations. Stations examined include East Croydon, Blackheath, St Mary Cray Station and West Dulwich stations.

Design Engineer – Stour River Footbridge Design works February - (£20k)

Carried out feasibility studies and produced option report. I modelled, analysed and checked capacity of preferred flow deck option taking into account sustainability.

❖ LAING O'ROURKE — (August 2003 – November 2004)

Carried out setting out and levelling works on Channel Tunnel Rail link Project and Terminal 5 Projects.

❖ NANA ARKONUAH LTD – Civil Engineering and Building Contractors — (January 2002 – June 2003)

Project Engineer / Environmental Manager

Project overview: Construction of 4 Units Two Storey (\$ 0.1m). Client: Government of Ghana. The project comprised the construction of two-unit classroom block and two unit dormitory for Aburi Training College.

Responsibilities & Achievements: I reviewed project drawings and the site environs and established a network benchmarks in relation to the works to reduce setting out times. I issued requests for information, technical queries and chased approvals. I also undertook valuation of works executed on a daily basis and produced monthly valuations for progress claims. I prepared variations and pursued claims comprising additional works,

dayworks, and delays and stoppages. As a Project Engineer, I was also responsible for work allocation to teams, site safety, toolbox talks and job briefings

❖ PW Ghana Ltd (Part of PW Overseas Ltd of Ireland) — (March 1999 – October 2001)

Project Engineer / Contractor Responsible Engineer - Heavy Civil Engineering and Mining Construction

Project Overview: Client- Various. Contracts involved heavy civil engineering construction, mine infrastructure, airport and bulk ore mining for gold mines.

Responsibilities & Achievements: I successfully supervised construction of heavy civil engineering works including a heap leach pad for the leaching of gold, haul roads and mine set-up infrastructure, airport runway extension and arrival building re-modelling. On the mining projects, carried out quality assurance of design/construction work, computed monthly volumes of mined ore, prepared monthly progress report and submitted claims for work done. I carried out rise and fall analysis for contract variation. I also prepared Environmental Impact Assessments (EIA) and Environmental Management Plans (EMP) and monitored and enforced the EMP and health & safety standards. I provided on-the-job training for graduate engineers and technicians. Projects worked on include; *Satellite Goldfields – Construction and Operational Phases, Resolute Amansie Mines – Operational Phase, and Kotaka International Airport Extension Project (\$77 million).*

❖ GHANA HIGHWAY AUTHORITY (GHA) — (1997 – 1999)

Civil Engineer

Overview: Highway Sector Projects. Client: Government of Ghana and Donor Agencies including the World Bank. Projects included design, supervision, condition assessment and contract procurement on behalf of the Government of Ghana for highway and road investment schemes.

Responsibilities & Achievements: I carried out inspections, designed and checked designs of highway structures including drainage, culverts, retaining walls and bridges. I inspected works done by contractors. I prepared EIA on roads construction work. I successfully led a road condition survey team to obtain data on road roughness to assist in planning and scheduling resurfacing and repair works. Projects worked on include; *Accra–Kumasi Road Rehabilitation Scheme, Road Condition Survey and Survey for Assessment, Environmental Impact Assessment of Road Projects in Ghana.*

❖ ARCHITECTURAL AND ENGINEERING SERVICE LIMITED (AES Ltd) — (1996 – 1997)

Assistant Engineer

Project Overview: Client – Supreme Court Of Ghana. The project comprised the design, preparation of tender for contractor selection and supervision of the building construction.

Responsibilities & Achievements: I successfully designed and provided detailed structural drawings (using AutoCAD software) including bending schedules. I prepared contract specifications for construction and prepared tender documents for contractor selection. During the design I liaised with the company Architects to make changes to structural members where deemed necessary.

❖ ACCRA TECHNICAL UNIVERSITY — (1996 – 1999)

Part-time Lecturer

I was a part time structural Engineering Lecturer for 2nd and final year students.

❖ IRRIGATION DEVELOPMENT AUTHORITY — (1996 – 1997)

Graduate Engineer

Client: Government of Ghana. The project involved the design of irrigation dam and drainage network to rice farms at Torlu in Ghana. I was part of the survey team that surveyed the catchment area. I estimate the water capacity of the farm and assisted in designing the dam.

EDUCATION & TRAINING

- Post-Graduate Diploma (PGDip) Actuarial Science – University of Leicester (2013)
- Master of Science (MSc) Structural Engineering – London Southbank University (2004)
- Master of Business Administration (MBA) International Management of Resources & Environment Freiberg University of Mining and Technology, Germany (2002)
- BSc (Bachelor of Science) Civil Engineering – University of Science and Technology, Ghana (1996)

LANGUAGES

Fluent English

HOBBIES AND INTEREST

- Football
- Rugby
- Music